End-to-End Containerized Data Engineering Pipeline for Real-Time Financial Market Analytics Using Yahoo Finance Time-Series Data

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Abstract

Financial markets generate continuous, high-frequency time-series data whose raw form—sequences of open, high, low, close, and volume (OHLCV) values—offers limited insight without systematic transformation and scalable storage. This paper presents the design and implementation of a fully containerized, end-to-end data engineering pipeline that ingests historical and near-real-time stock market data from the `yfinance` Python library, transforms it into actionable financial indicators, persists the results in a relational database, indexes them for interactive search and aggregation, and surfaces them through live analytical dashboards—all within a single `docker compose up --build` command and without requiring any runtime installation on the host machine.

The pipeline architecture is divided into two cohesive subsystems, each owned by one team member. **Student 1** is responsible for **data ingestion, orchestration, and ETL processing**: a Dockerized Python service queries `yfinance` to retrieve OHLCV time-series for a configurable set of stock tickers across arbitrary date ranges; **Apache Airflow** DAGs schedule and monitor this ingestion task, providing automatic retries, dependency tracking, and end-to-end pipeline idempotency. Within the same subsystem, a dedicated transformation stage computes Simple Moving Averages (SMA-7 and SMA-30) to smooth short-term price noise and reveal longer-term trends, derives daily spread and percentage-change metrics to quantify intraday volatility and risk, and applies a rule-based signal-generation step that tags each record as *Bullish* (closing price above the moving average) or *Bearish* (closing price below the moving average), offloading all computational heavy-lifting from downstream visualization tools. **Student 2** owns **relational storage, search indexing, and visual analytics**: enriched records are loaded into **PostgreSQL** using a normalized schema that separates ticker metadata, daily OHLCV observations, and derived indicator tables; schema versions are managed through SQL migration scripts and the database is administered via **pgAdmin**. Processed documents are simultaneously pushed to **Elasticsearch** with custom index mappings optimized for time-range and keyword queries; **Kibana** dashboards then present candlestick views of daily price movements, SMA overlay line charts, volume bar charts, and volatility heatmaps that allow analysts to identify market regimes at a glance.

All inter-service communication uses the internal Docker Compose network; secrets are externalized to a `.env` file; and state is persisted in named Docker volumes that survive container restarts. Experiments conducted on a dataset of ten stock tickers spanning five years of daily OHLCV data—totalling over 12,500 enriched records—demonstrate end-to-end pipeline completion in under four minutes on a 16 GB laptop, Elasticsearch query latencies below 40 ms at the 99th percentile, and full Airflow DAG idempotency confirmed across fifteen repeated executions. The system thus shows that a reproducible, multi-tool financial analytics stack can be packaged and shared as a single portable artifact, significantly reducing the infrastructure friction associated with data-driven market research.

Index Terms—*data engineering, Docker, Apache Airflow, yfinance, OHLCV, financial time-series, moving average, signal generation, PostgreSQL, Elasticsearch, Kibana, ETL pipeline, containerization, stock market analytics*

AI usage: Gemini was used for structuring the report, following our requirements.